

Project Executable Files

Date	15 March 2026
Team ID	
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	3 Marks

Project Executable Files

This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	NA
5	Environment configuration file (.env.example or similar)	NA
6	Deployed application URL (if hosted)	NA
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	NA
9	Test files and test results	Yes
10	Demo video or walkthrough (if required)	NA

Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below:

```

/VBCUA-project-root
/app
  main.py           - Streamlit application entry point
  transcription.py  - OpenAI Whisper speech-to-text module
  semantic_analysis.py - Sentence-BERT semantic similarity scoring
  audio_features.py - Librosa/pYIN based fluency & pause analysis
  report_generator.py - ReportLab PDF report builder
  utils.py          - Session state & scoring helper functions
/assets             - Reference concept dataset, sample audio, logos
/tests             - Unit tests for transcription, similarity & audio
modules
/web-version
  index.html        - Standalone HTML/JS version (Web Speech API,
                     Web Audio API, jsPDF)
requirements.txt    - Python dependencies
README.md           - Setup and run instructions
  
```

Step 3: Deployment / Access Details

Hosted / Deployed URL	Not hosted – runs locally via Streamlit (streamlit run app/main.py)
Login Credentials (Demo)	NA – no authentication required
Platform / Hosting Provider	NA – local execution (Streamlit)
Repository Link	<i>[Add GitHub repository link here]</i>
Demo Video Link	<i>[Add demo video link here]</i>

Step 4: Run Instructions

Provide clear, step-by-step instructions for the evaluator to run your project locally:

Pre-requisites:
 Python 3.9 or higher, pip, ffmpeg installed on the system (required for
 Whisper audio decoding), ~4 GB RAM recommended for the Whisper base model.

Step 1: Clone the repository: `git clone <repository-url>`
 Step 2: Navigate to the project folder: `cd VBCUA-project-root`
 Step 3: Install dependencies: `pip install -r requirements.txt`
 Step 4: Run the application: `streamlit run app/main.py`
 Step 5: Open the browser at: `http://localhost:8501`
 Step 6: Record or upload a spoken explanation, select the target concept,
 and click "Analyze" to view the score and download the PDF report.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Whisper transcription accuracy may drop with heavy background noise or strong accents	Ask users to record in a quiet environment; upgrading to the Whisper "medium" model improves accuracy
2	Longer audio recordings increase processing time	Recommend keeping explanations under 2 minutes for optimal response time
3	First run takes longer while Whisper and Sentence-BERT models load/download	Expected behaviour; a loading spinner is shown to the user